



IIT ROORKEE



NPTEL ONLINE
CERTIFICATION COURSE

Human Resource Analytics

Lecture 04

Pitfalls of data science framework and Basics of data visualization

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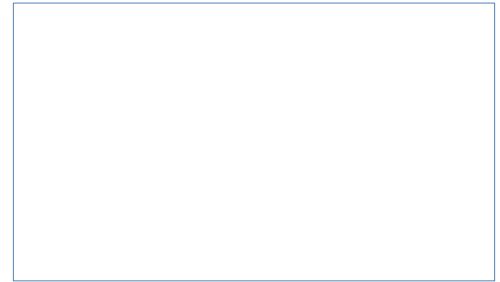
Faculty-Human Resource Management

XLRI-Jamshedpur



CONTENT

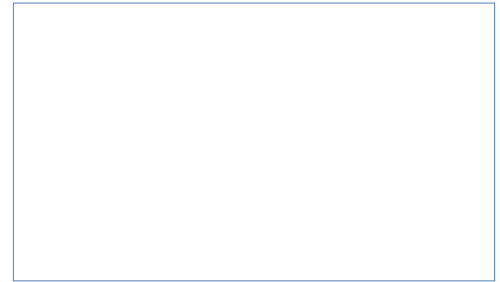
- ☐ Total employee salary structure (Quartile-wise) (Pay-Gap Analysis)
- ☐ Annual Salary
- ☐ Analysis of employee benefits
- ☐ Top Management compensation
- ☐ Behavior encouragement Plan Matrix
- ☐ Incentive Plan Planning



Pitfalls of data science framework and Basics of data visualization

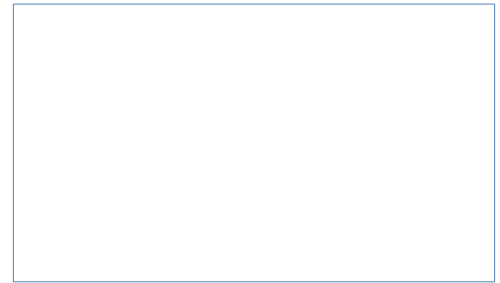
- **Confirmation Trap**

When we pay more attention to findings that align with our prior beliefs, and ignore other facts and patterns in the data, we fall into the confirmation.



- **Overconfidence trap**

Senior decision makers who have been promoted based on past successes are especially to this bias, since they have received positive signals about their decision-making abilities throughout their career. It prevents us from questioning our methods, motivation and the way we communicate our findings to others.

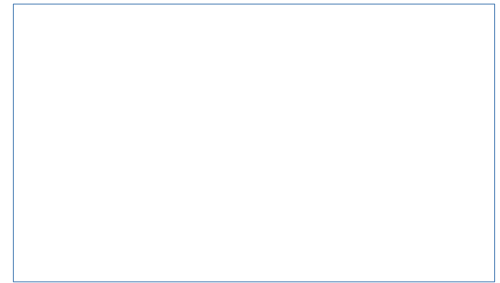


- **Over fitting trap**

When your model yields surprising or counterintuitive predictions, you have made an exciting new discovery or it may be the result of over fitting.

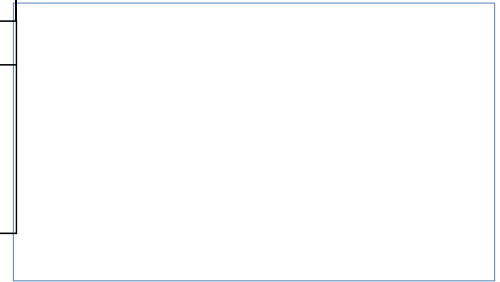
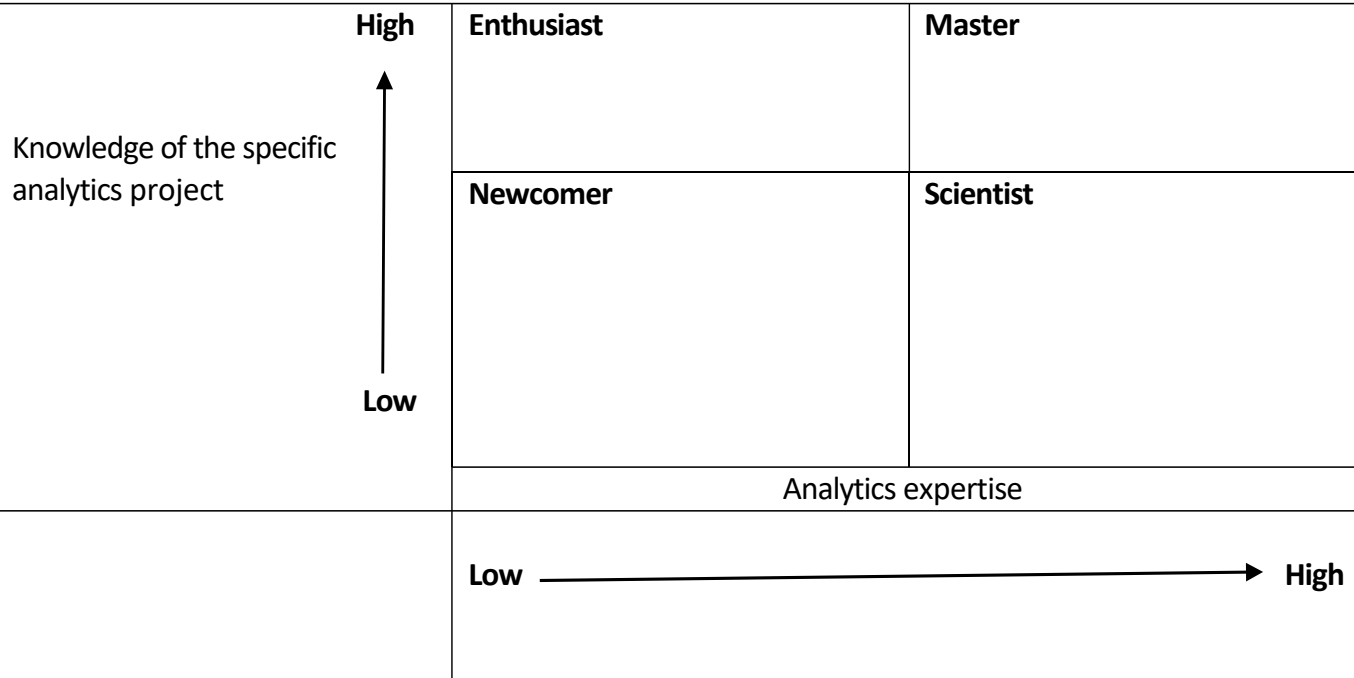
Do not let your analytics hide the truth

Pay attention to outlier in data



Basics of data visualization

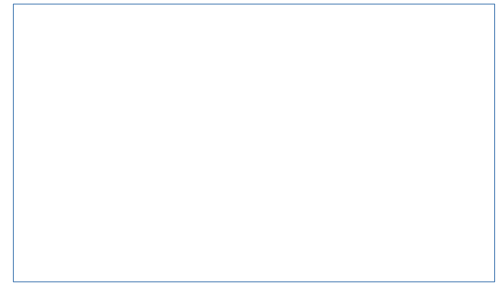
Know your audience



- Who is the audience, and how will it read and interpret the information?
- Can you assume these individuals have knowledge of the terminology and concepts you will use?
- Or do you need to guide them with clues in the visualization

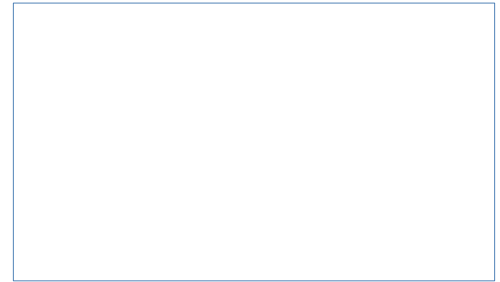
Note – An audience of expert will have different expectations than a general audience

What is the visualization's functional role, and how can viewers take action from it?



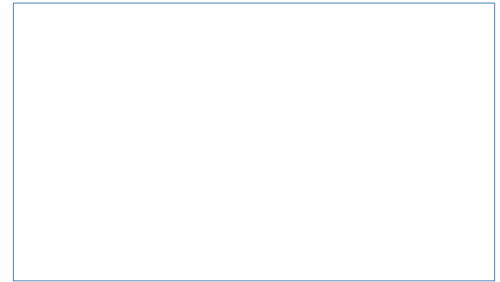
Understand the aim of data visualization

- Confirmation
- Education
- Exploration



Ask questions after making the graph

- What message am I trying to convey
- Do my visuals accurately reflect the numbers?
- Is my data memorable?



Thank You

